CHANDUBHAI S. PATEL INSTITUTE OF TECHNOLOGY (CSPIT)

Department of Computer Science & Engineering • CHARUSAT

Campusly (StudentHub) — Final Project Overview

Student Name: Jay Navadiya

Student ID / Roll No: 25CS052

Course: ITUE203 - Web Development Frameworks

Semester / Branch: B.Tech CSE (Div-I)

GitHub Repo: github.com/cs-cspit/Student-Portal-25CS052

Database Engine: MySQL 8.x (3NF Relational Database `studenthub`)

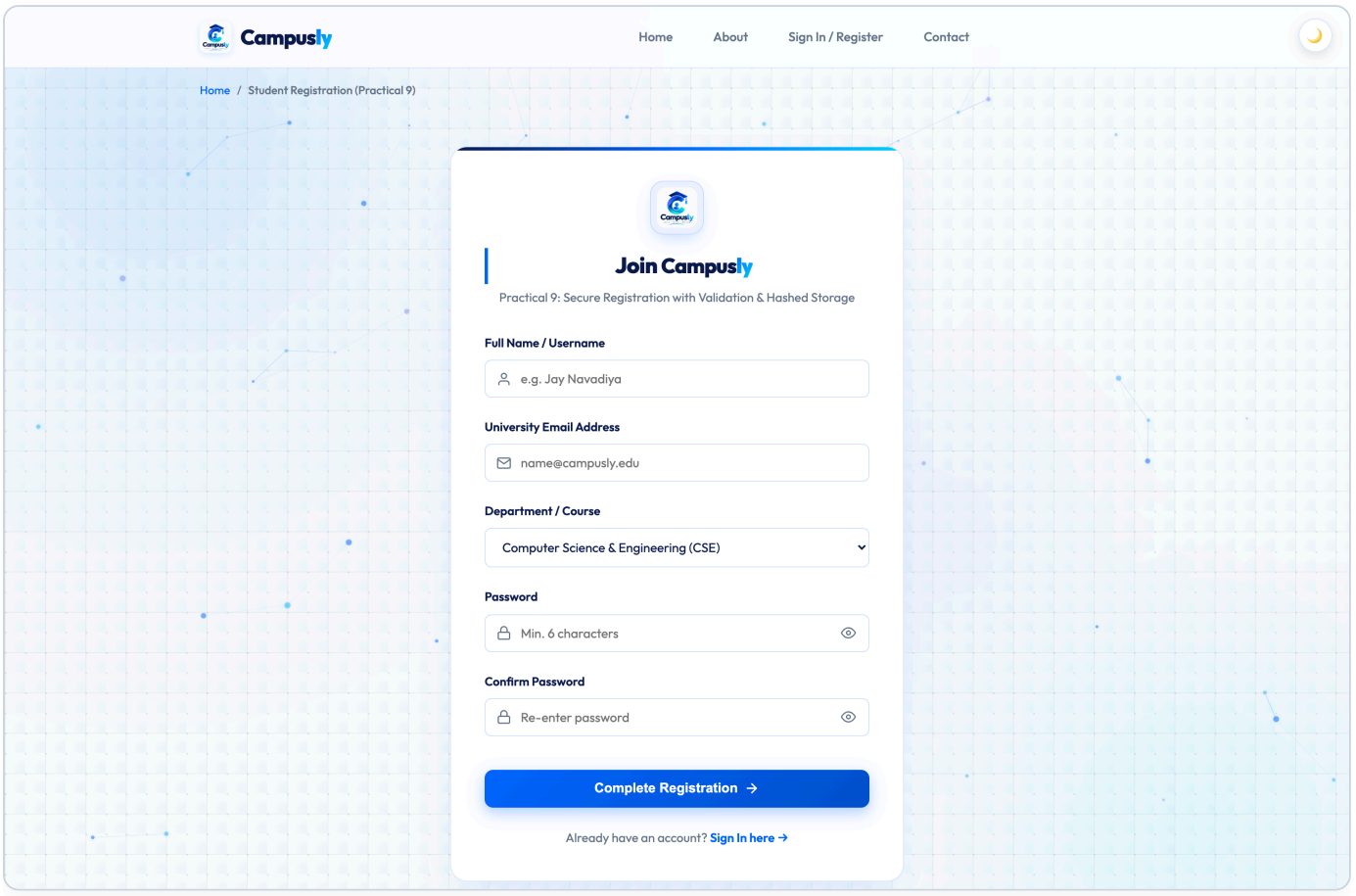
Executive Requirements & Features Satisfaction Matrix

Module / Requirement	Features Implemented & Demonstrated	Verification Proof	Status
User Registration	Full form validation, bcrypt password hashing, dual relational insertion (students + users)	Figure 1.1 - 1.4	100% SATISFIED
Session Inactivity Timeout	10-minute zero-drift epoch timer, 60s warning modal, stay logged in extension & auto-redirect	Figure 2.1 - 2.2	100% SATISFIED
Authentication & RBAC	Bcrypt verification, session fixation prevention, student/admin dynamic navigation, HTTP 403 barrier	Figure 3.1 - 3.4	100% SATISFIED
Executive Admin Dashboard	Dynamic relational statistics, monthly chart, user active/inactive toggle, relational audit logging	Figure 4.1 - 4.2	100% SATISFIED
Student Portal & Schedules	Personalized student dashboard, weekly timetable schedule, academic study materials repository	Figure 5.1 - 5.3	100% SATISFIED
Student Directory & AJAX CRUD	Asynchronous add modal, debounced live search, course filters, instant CSV export	Figure 6.1 - 6.3	100% SATISFIED
Campus Events Catalog	Card visualizer, poster validation, student event registration junction linking tables in 3NF	Figure 7.1	100% SATISFIED
Responsive Mobile Viewport	Mobile layout testing on 390px (iPhone 14 Pro), accessible touch targets, collapsible menu	Figure 8.1	100% SATISFIED
Local & Apache Deployment	Apache Virtual Host configuration (studenthub.local), MySQL database dump (studenthub.sql), Git origin	Figure 9.1 - 9.3	100% SATISFIED

1. User Registration & Dual-Layer Validation

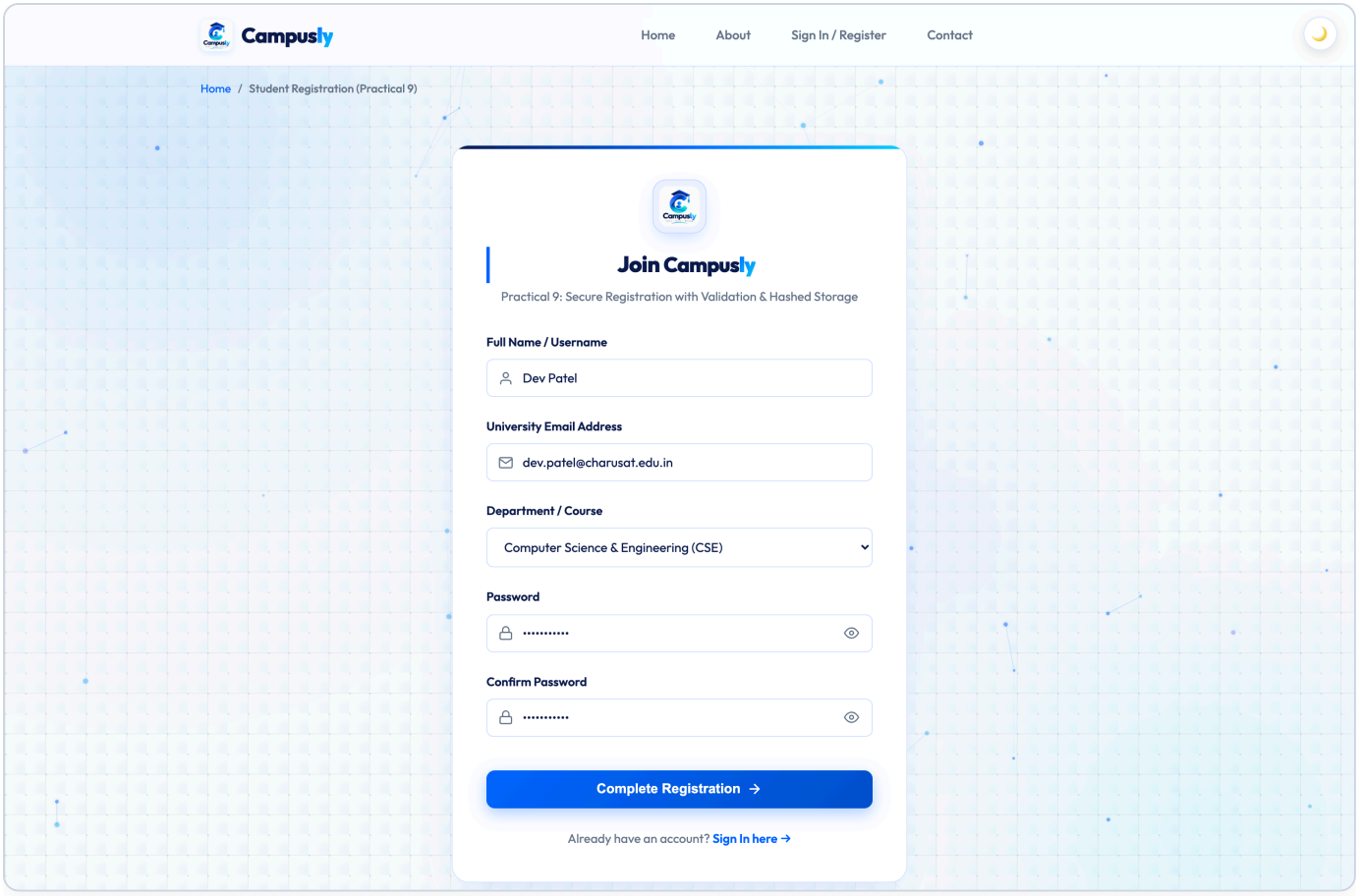
1.1 Student Registration Form Interface

Initial clean state of the student registration gateway (register.php) featuring input fields for Full Name, University Email, Department, Password, and Password Confirmation in portal light mode.



1.2 Registration Process: Form Filled with Student Data

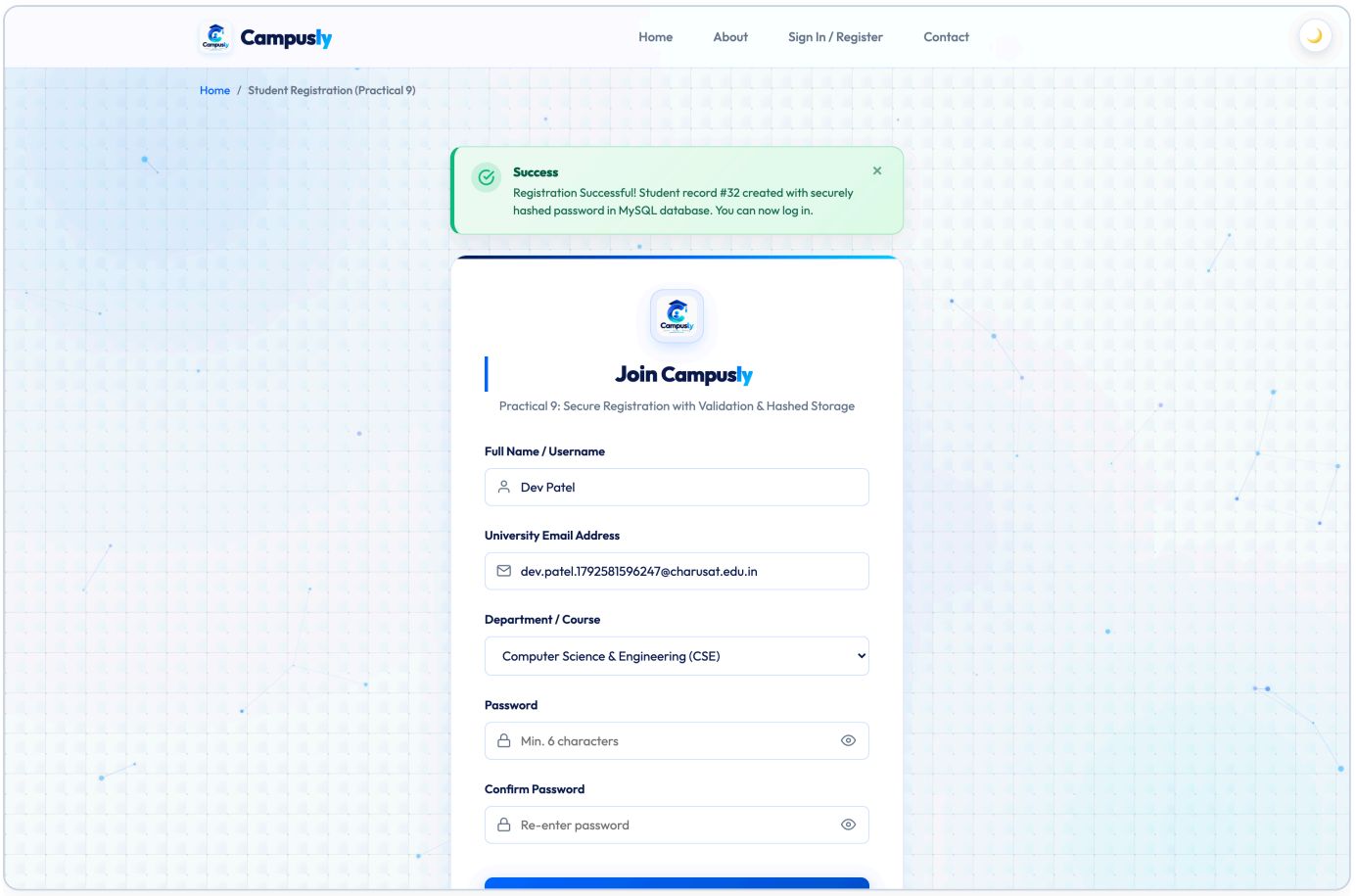
Registration form actively populated with student details (Dev Patel, CSE Department, credentials) demonstrating client-side validation checks before submission.


Campusly
Home
About
Sign In / Register
Contact
Home / Student Registration (Practical 9)
Join Campusly
Practical 9: Secure Registration with Validation & Hashed Storage
Full Name / Username
Dev Patel
University Email Address
dev.patel@charusat.edu.in
Department / Course
Computer Science & Engineering (CSE)
Password
Confirm Password
Complete Registration →
Already have an account? Sign In here →

1. User Registration & Dual-Layer Validation (Contd.)

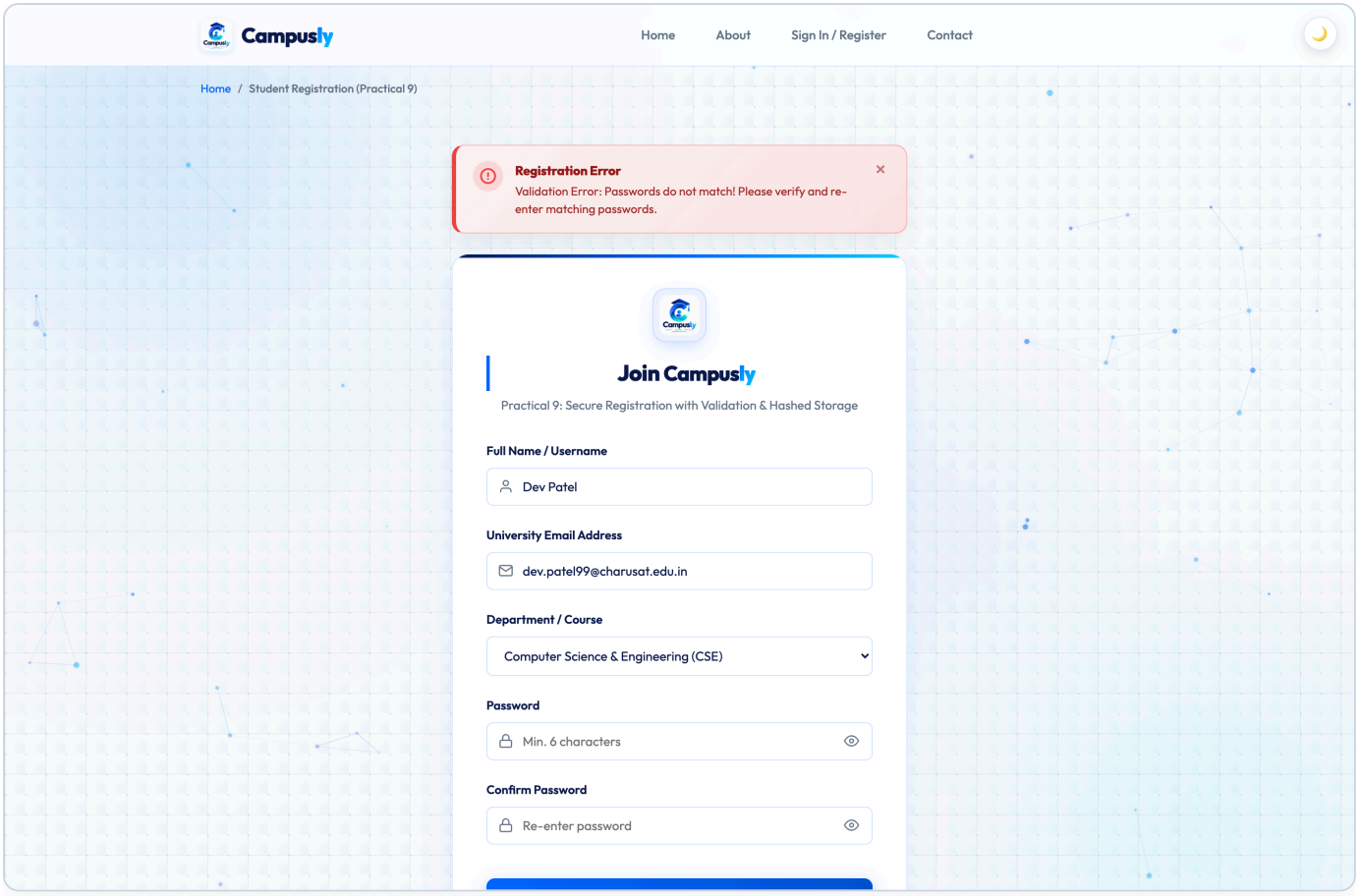
1.3 Successful User Registration & Relational MySQL Insertion

Visual confirmation of successful registration: A green notification alert confirms student record creation with securely hashed bcrypt password in the MySQL database (students and users tables).



1.4 Server-Side Validation Error Handling (Password Mismatch)

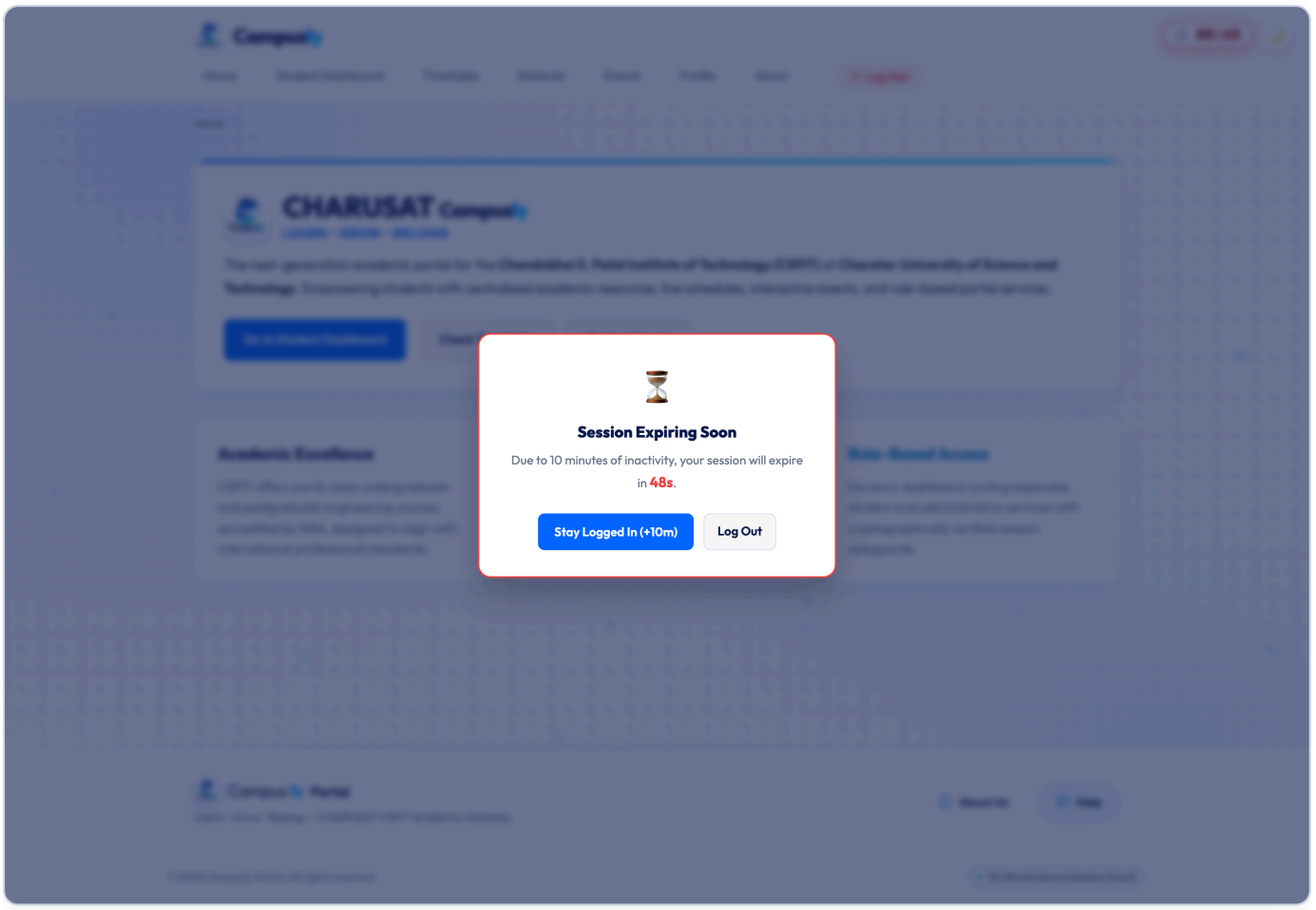
Demonstration of input validation safeguards: When password fields do not match, the portal displays a prominent validation notification preventing dirty data insertion.


Campusly
Home
About
Sign In / Register
Contact
Home / Student Registration (Practical 9)
Registration Error
Validation Error: Passwords do not match! Please verify and re-enter matching passwords.
Join Campusly
Practical 9: Secure Registration with Validation & Hashed Storage
Full Name / Username
Dev Patel
University Email Address
dev.patel99@charusat.edu.in
Department / Course
Computer Science & Engineering (CSE)
Password
Min. 6 characters
Confirm Password
Re-enter password

2. 10-Minute Zero-Drift Session Security & Inactivity Timeout

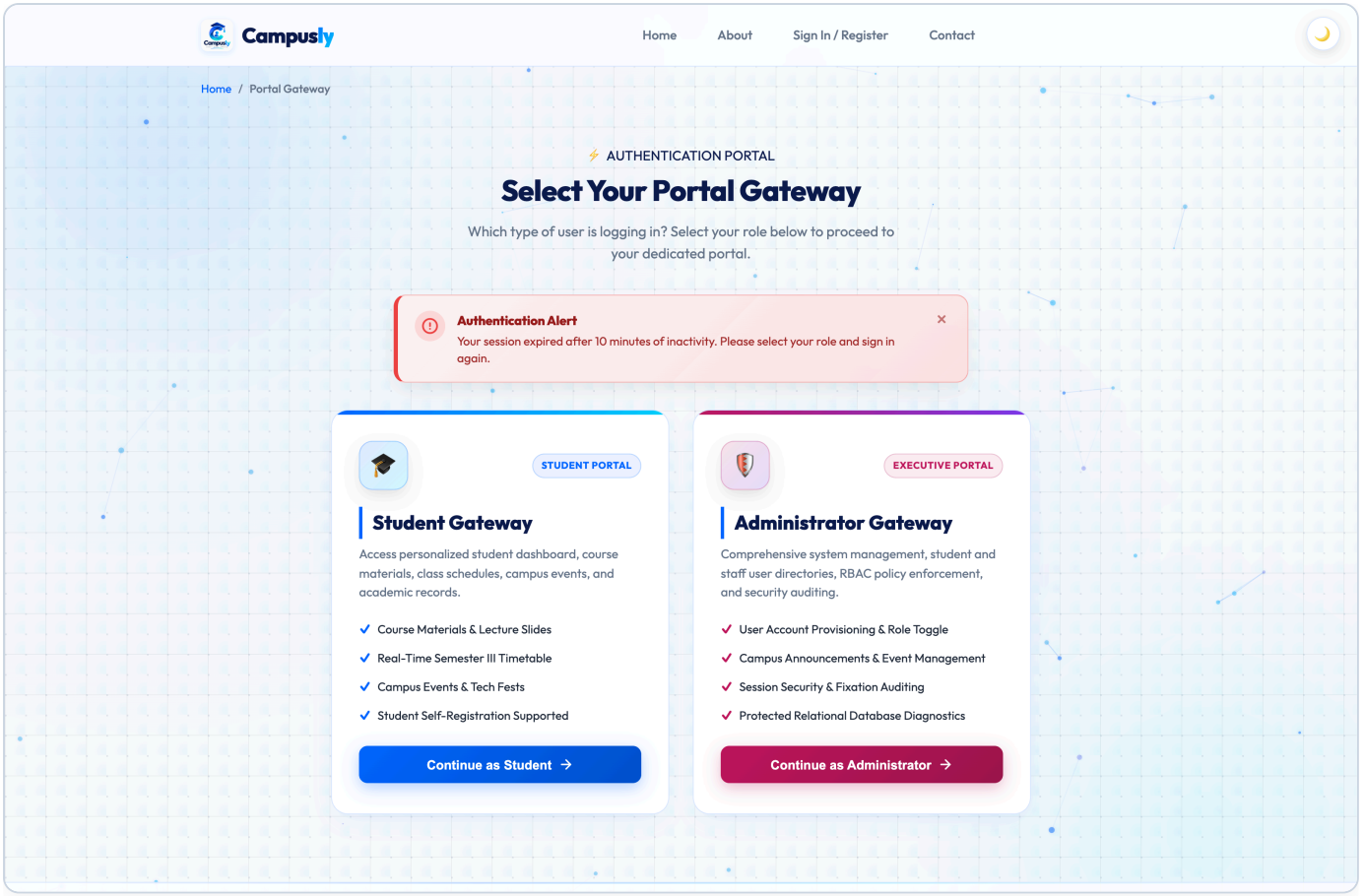
2.1 Session Timeout Warning Modal (Countdown Timer & Extension Action)

After 9 minutes of user inactivity, the portal triggers a non-intrusive warning modal displaying an active countdown (48s). Users can click "Stay Logged In (+10m)" to extend their session or "Log Out" to immediately terminate.



2.2 Session Invalidation & Automatic Redirect to Sign-In

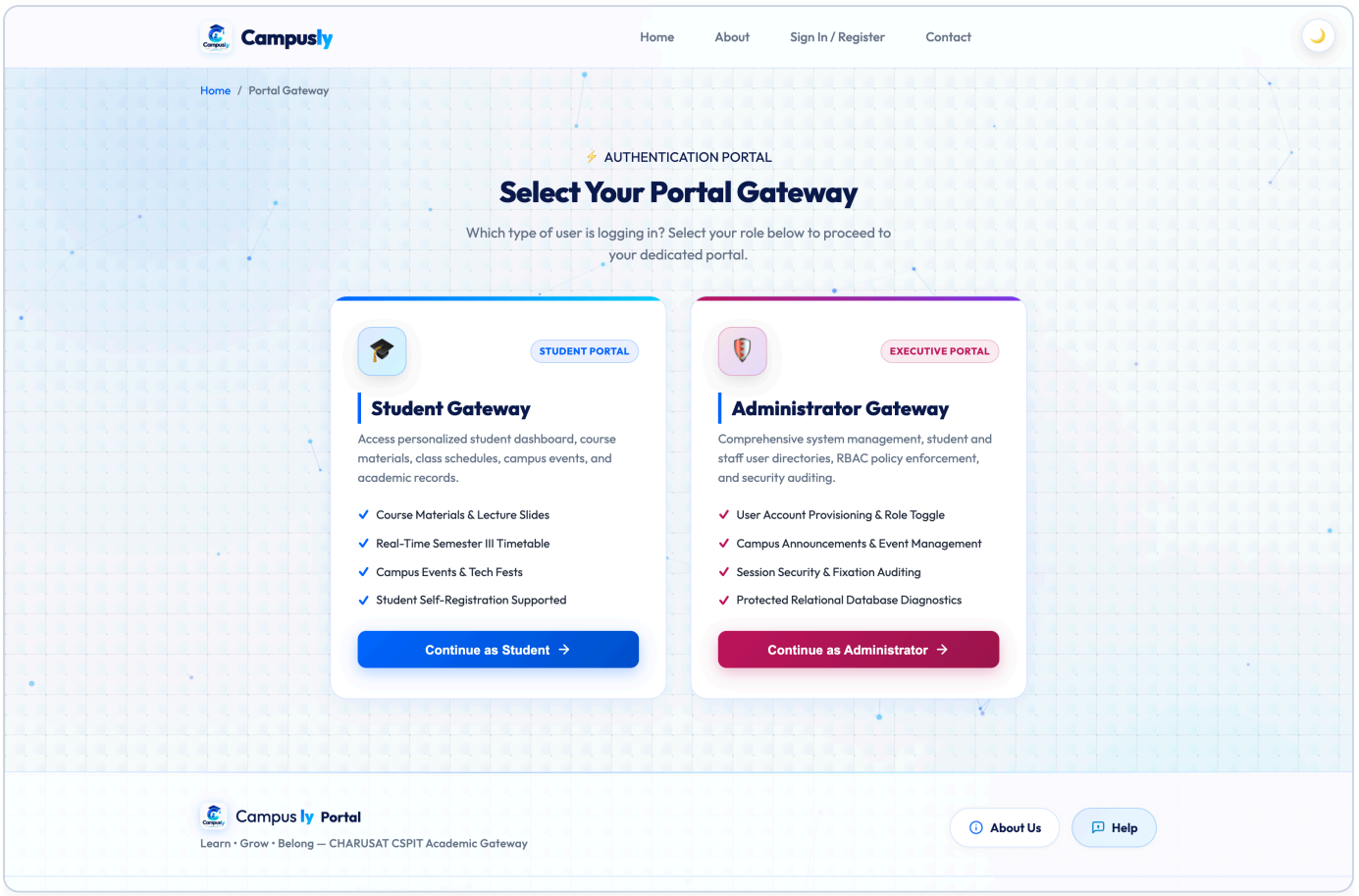
Upon complete session expiration (10 minutes), the server-side epoch guard invalidates the session cookie and redirects the user to login.php?msg=timeout with a clear notification informing them of session termination.


Campusly
Home
About
Sign In / Register
Contact
Home / Portal Gateway
AUTHENTICATION PORTAL
Select Your Portal Gateway
Which type of user is logging in? Select your role below to proceed to your dedicated portal.
Authentication Alert
Your session expired after 10 minutes of inactivity. Please select your role and sign in again.
STUDENT PORTAL
Student Gateway
Access personalized student dashboard, course materials, class schedules, campus events, and academic records.
Course Materials & Lecture Slides
Real-Time Semester III Timetable
Campus Events & Tech Fests
Student Self-Registration Supported
Continue as Student
EXECUTIVE PORTAL
Administrator Gateway
Comprehensive system management, student and staff user directories, RBAC policy enforcement, and security auditing.
User Account Provisioning & Role Toggle
Campus Announcements & Event Management
Session Security & Fixation Auditing
Protected Relational Database Diagnostics
Continue as Administrator

3. Authentication Gateway & Role-Based Access Control (RBAC)

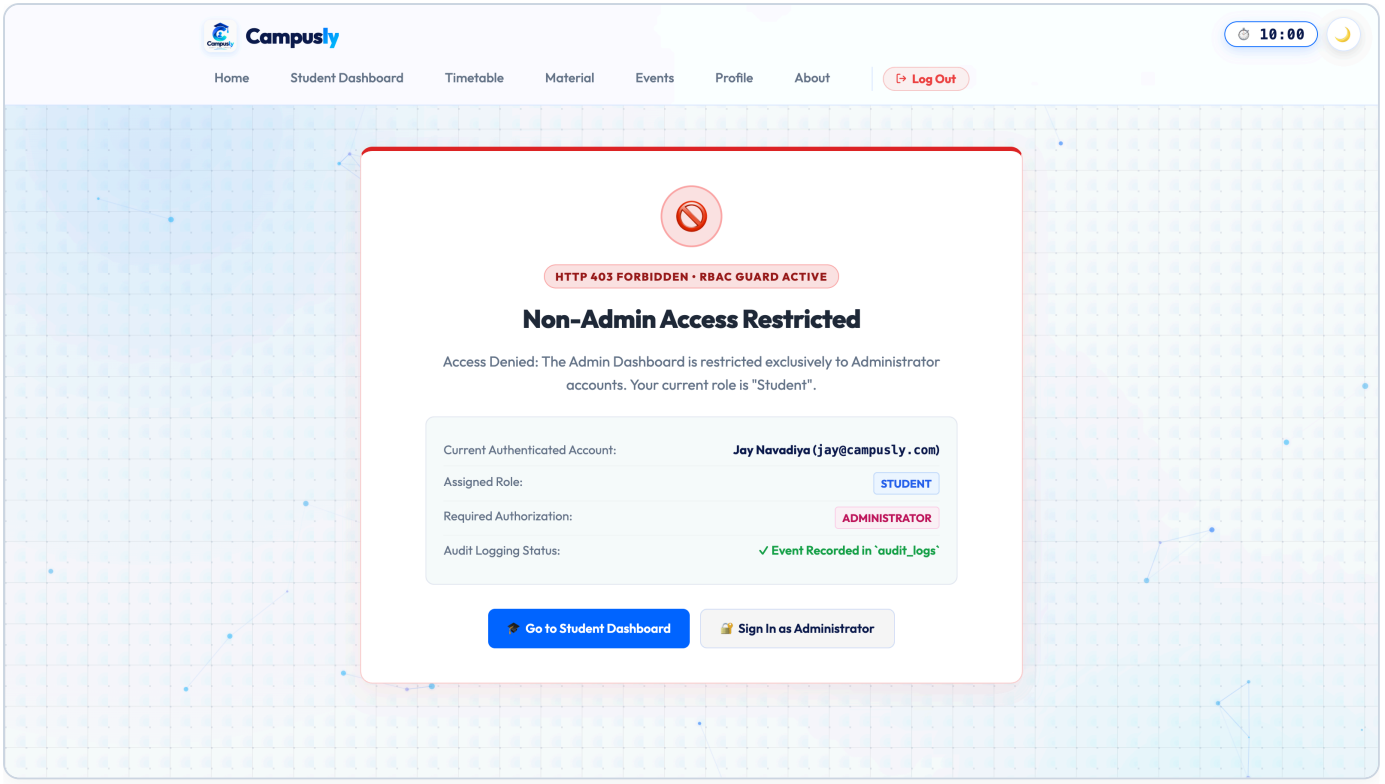
3.1 Authentication Gateway & Dual-Role Switcher

Unified login portal (login.php) with instant role-switching pills (Student and Administrator), demo account prefill pills, remember-me token option, and clean light-themed inputs.



3.2 Role-Based Access Control Interception (HTTP 403 Forbidden Alert)

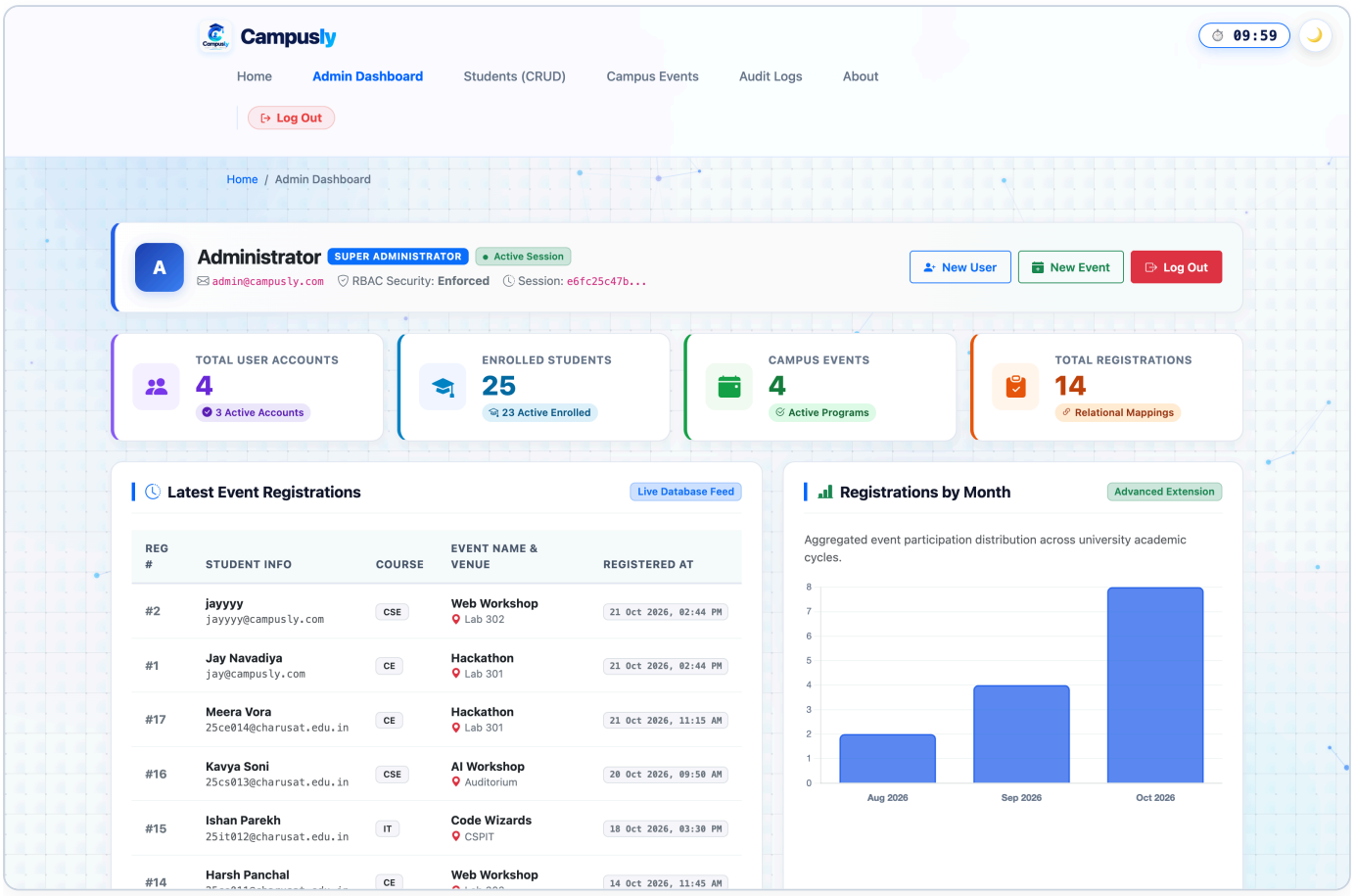
Strict security barrier: When an authenticated Student (Jay Navadiya) attempts to access restricted Admin routes (/dashboard.php), the system intercepts the request, blocks execution with HTTP 403, and records an audit log.


Campusly
10:00
Home
Student Dashboard
Timetable
Material
Events
Profile
About
Log Out
HTTP 403 FORBIDDEN · RBAC GUARD ACTIVE
Non-Admin Access Restricted
Access Denied: The Admin Dashboard is restricted exclusively to Administrator accounts. Your current role is "Student".
Current Authenticated Account:
Jay Navadiya (jay@campusly.com)
Assigned Role:
STUDENT
Required Authorization:
ADMINISTRATOR
Audit Logging Status:
✓ Event Recorded in `audit_logs`
Go to Student Dashboard
Sign In as Administrator

4. Executive Admin Control Center, Analytics & Audit Logging

4.1 Executive Admin Dashboard with Dynamic Statistics & Registration Chart

Administrator overview (dashboard.php) featuring live database KPIs (4 Users, 25 Enrolled Students, 4 Events, 14 Registrations), latest relational registration feed, and Chart.js monthly analytics.



4.2 Relational Security Audit Trail (audit_logs Table Verification)

Full traceability log tracking administrative actions and security violations (LOGIN_SUCCESS, USER_STATUS_TOGGLE, UNAUTHORIZED_ACCESS_403, EVENT_CREATE) with IP addresses and timestamps.

Audit Log Trail (`audit_logs`)

Complete traceability record showing all administrative updates, logins, status toggles, and security violations.

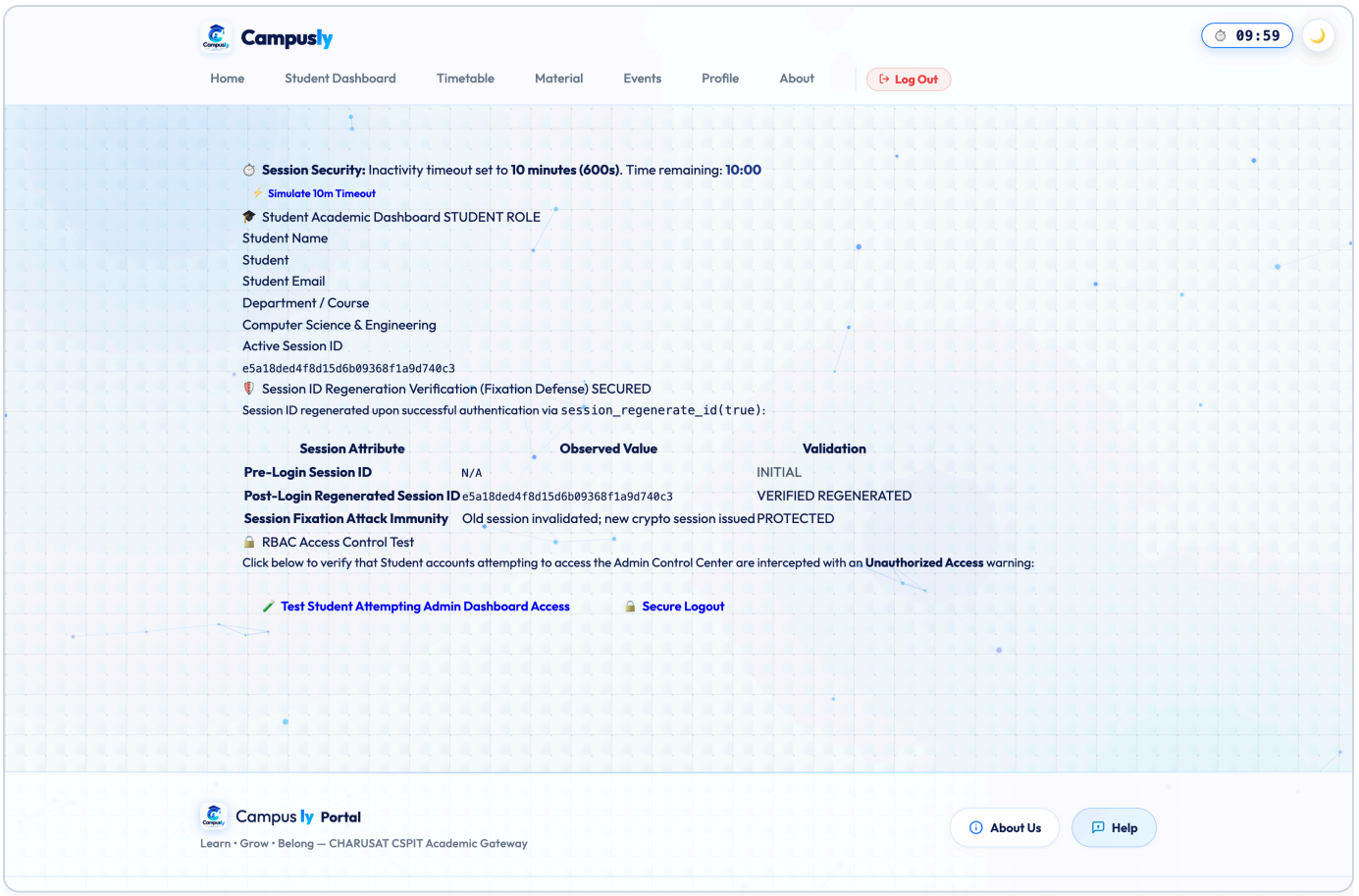
26 Total Recorded Events

LOG ID	TIMESTAMP	USER & ROLE	ACTION TYPE	ACTION DETAILS	IP ADDRESS
#26	21 Oct 2026, 04:34:28 PM	admin@campusly.com ADMIN	LOGIN_SUCCESS	Admin user admin@campusly.com authenticated successfully	127.0.0.1
#25	21 Oct 2026, 04:30:53 PM	jay@campusly.com STUDENT	UNAUTHORIZED_ACCESS_403	Forbidden access attempt to /dashboard.php by role: student (jay@campusly.com)	127.0.0.1
#24	21 Oct 2026, 04:30:52 PM	jay@campusly.com STUDENT	UNAUTHORIZED_ACCESS_403	Forbidden access attempt to /dashboard.php by role: student (jay@campusly.com)	127.0.0.1
#23	21 Oct 2026, 04:30:45 PM	jay@campusly.com STUDENT	LOGIN_SUCCESS	Student user jay@campusly.com authenticated successfully	127.0.0.1
#22	21 Oct 2026, 04:30:37 PM	admin@campusly.com ADMIN	LOGIN_SUCCESS	Admin user admin@campusly.com authenticated successfully	127.0.0.1
#21	21 Oct 2026, 03:20:08 PM	jay@campusly.com STUDENT	UNAUTHORIZED_ACCESS_403	Forbidden access attempt to /dashboard.php by role: student (jay@campusly.com)	127.0.0.1
#20	21 Oct 2026, 03:20:07 PM	jay@campusly.com STUDENT	LOGIN_SUCCESS	Student user jay@campusly.com authenticated successfully	127.0.0.1
#19	21 Oct 2026, 03:19:57 PM	admin@campusly.com ADMIN	LOGIN_SUCCESS	Admin user admin@campusly.com authenticated successfully	127.0.0.1

5. Student Academic Portal, Schedule & Study Materials

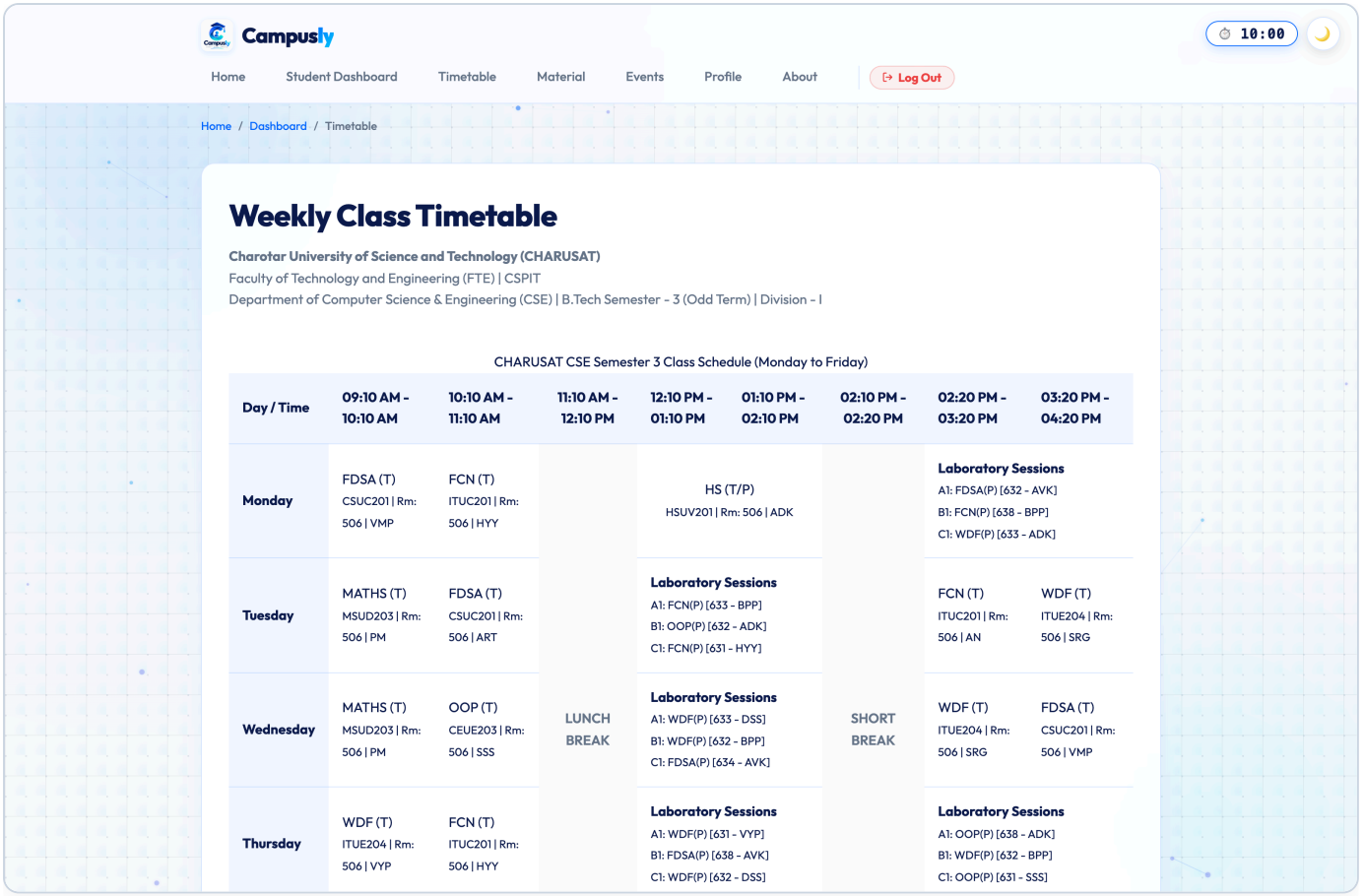
5.1 Personalized Student Academic Dashboard (student_dashboard.php)

Personalized student view for Jay Navadiya (B.Tech Computer Engineering) displaying academic status, registered event badges, and quick links to course resources.



5.2 Academic Class Timetable & Lab Allocations

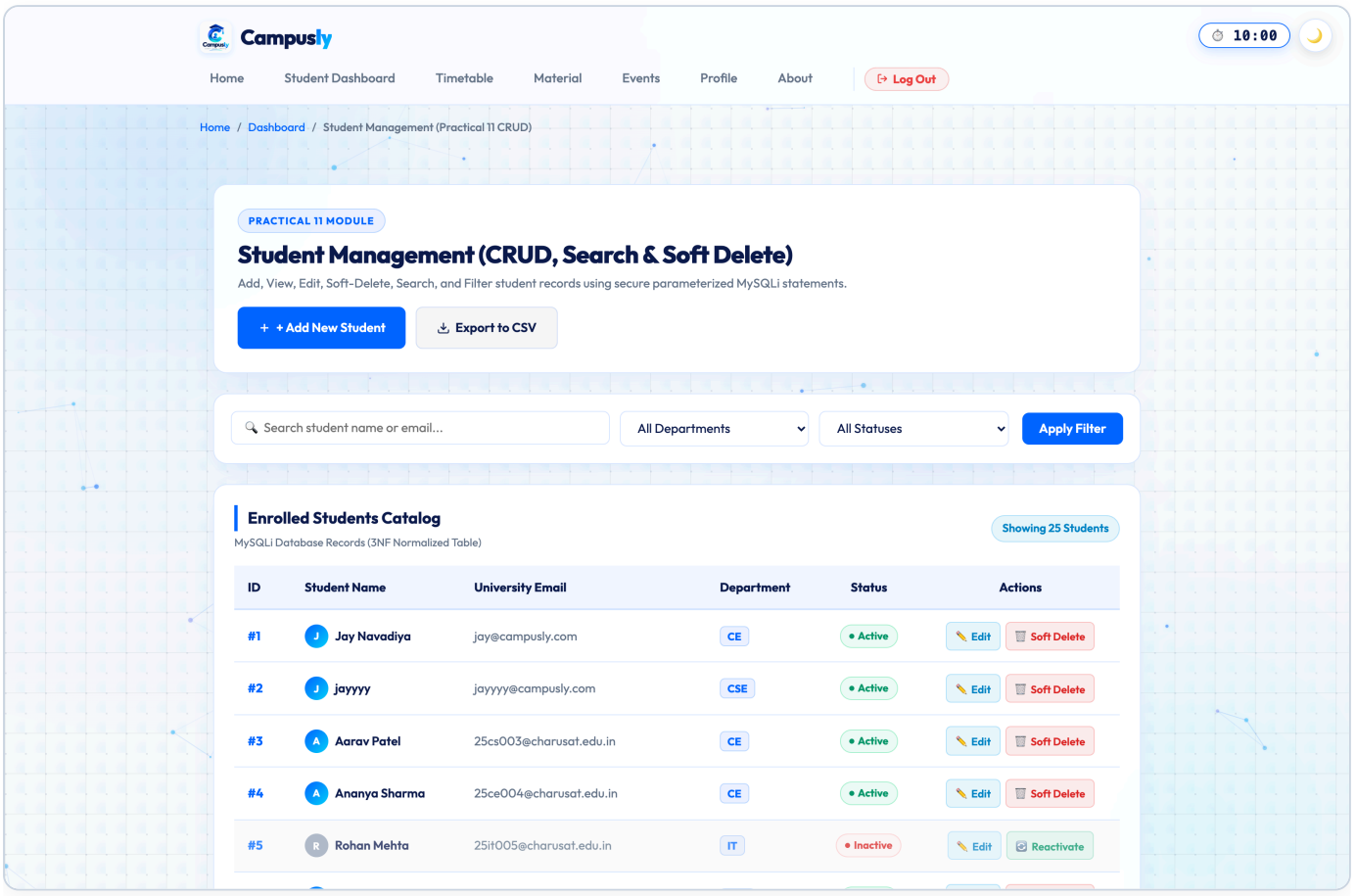
Weekly class timetable (timetable.php) displaying time slots, course codes (ITUE203, CE204), faculty names, and designated laboratory numbers.


Campusly
10:00
Home
Student Dashboard
Timetable
Material
Events
Profile
About
Log Out
Home / Dashboard / Timetable
Weekly Class Timetable
Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE) | CSPIT
Department of Computer Science & Engineering (CSE) | B.Tech Semester - 3 (Odd Term) | Division - I
CHARUSAT CSE Semester 3 Class Schedule (Monday to Friday)
Day / Time
09:10 AM - 10:10 AM
10:10 AM - 11:10 AM
11:10 AM - 12:10 PM
12:10 PM - 01:10 PM
01:10 PM - 02:10 PM
02:10 PM - 02:20 PM
02:20 PM - 03:20 PM
03:20 PM - 04:20 PM
Monday
FDSA (T) CSUC201 | Rm: 506 | VMP
FCN (T) ITUC201 | Rm: 506 | HYY
HS (T/P) HSUV201 | Rm: 506 | ADK
Laboratory Sessions A1: FDSA(P) [632 - AVK] B1: FCN(P) [638 - BPP] C1: WDF(P) [633 - ADK]
Tuesday
MATHS (T) MSUD203 | Rm: 506 | PM
FDSA (T) CSUC201 | Rm: 506 | ART
Laboratory Sessions A1: FCN(P) [633 - BPP] B1: OOP(P) [632 - ADK] C1: FCN(P) [631 - HYY]
FCN (T) ITUC201 | Rm: 506 | AN
WDF (T) ITUE204 | Rm: 506 | SRG
Wednesday
MATHS (T) MSUD203 | Rm: 506 | PM
OOP (T) CEUE203 | Rm: 506 | SSS
LUNCH BREAK
Laboratory Sessions A1: WDF(P) [633 - DSS] B1: WDF(P) [632 - BPP] C1: FDSA(P) [634 - AVK]
SHORT BREAK
WDF (T) ITUE204 | Rm: 506 | SRG
FDSA (T) CSUC201 | Rm: 506 | VMP
Thursday
WDF (T) ITUE204 | Rm: 506 | VYP
FCN (T) ITUC201 | Rm: 506 | HYY
Laboratory Sessions A1: WDF(P) [631 - VYP] B1: FDSA(P) [638 - AVK] C1: WDF(P) [632 - DSS]
Laboratory Sessions A1: OOP(P) [638 - ADK] B1: WDF(P) [632 - BPP] C1: OOP(P) [631 - SSS]

6. Student Directory Management & Asynchronous AJAX CRUD

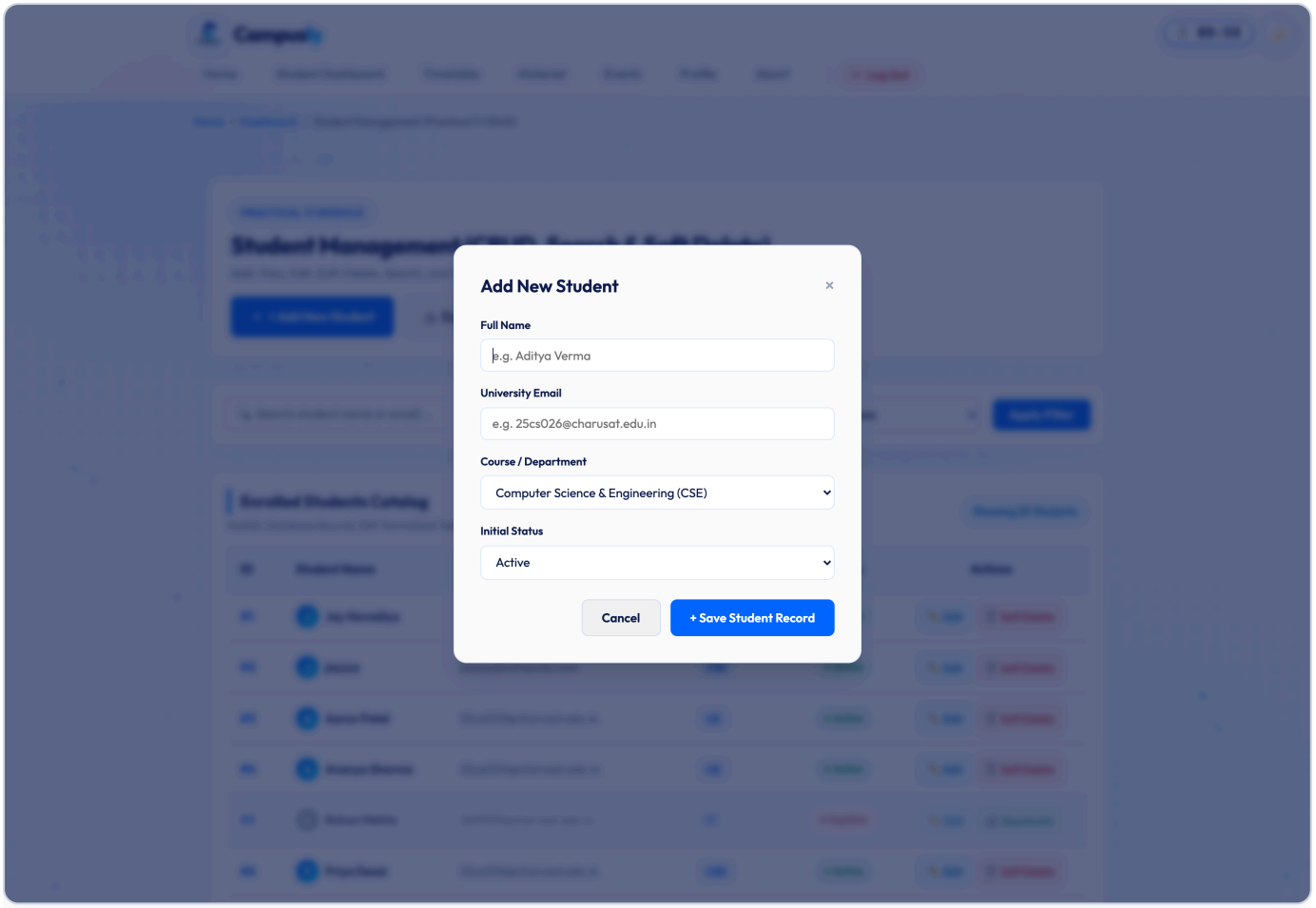
6.1 Enrolled Students Directory Interface with Department Badges

Student management directory (students.php) showing 3NF relational records with Department badges (CSE, CE, IT, EC), status badges, and action buttons.



6.2 Asynchronous AJAX Add Student Modal Dialog

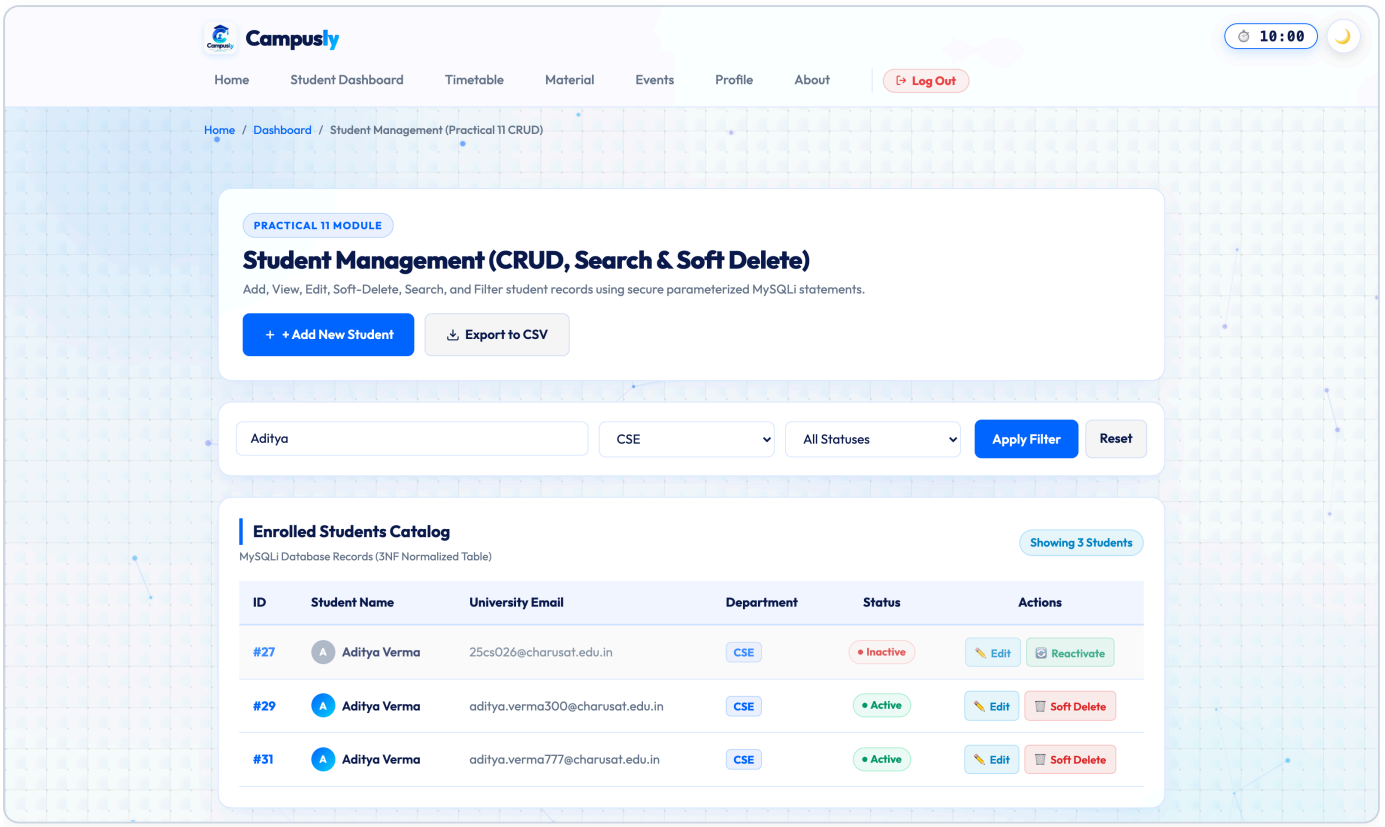
AJAX Add Student modal dialog open in light theme mode, enabling seamless record insertion without reloading the page.


Add New Student
Full Name
e.g. Aditya Verma
University Email
e.g. 25cs026@charusat.edu.in
Course / Department
Computer Science & Engineering (CSE)
Initial Status
Active
Cancel
+ Save Student Record

6. Student Directory Management & Asynchronous AJAX CRUD (Contd.)

6.3 Live Keystroke Debounced Search & Department Filtering

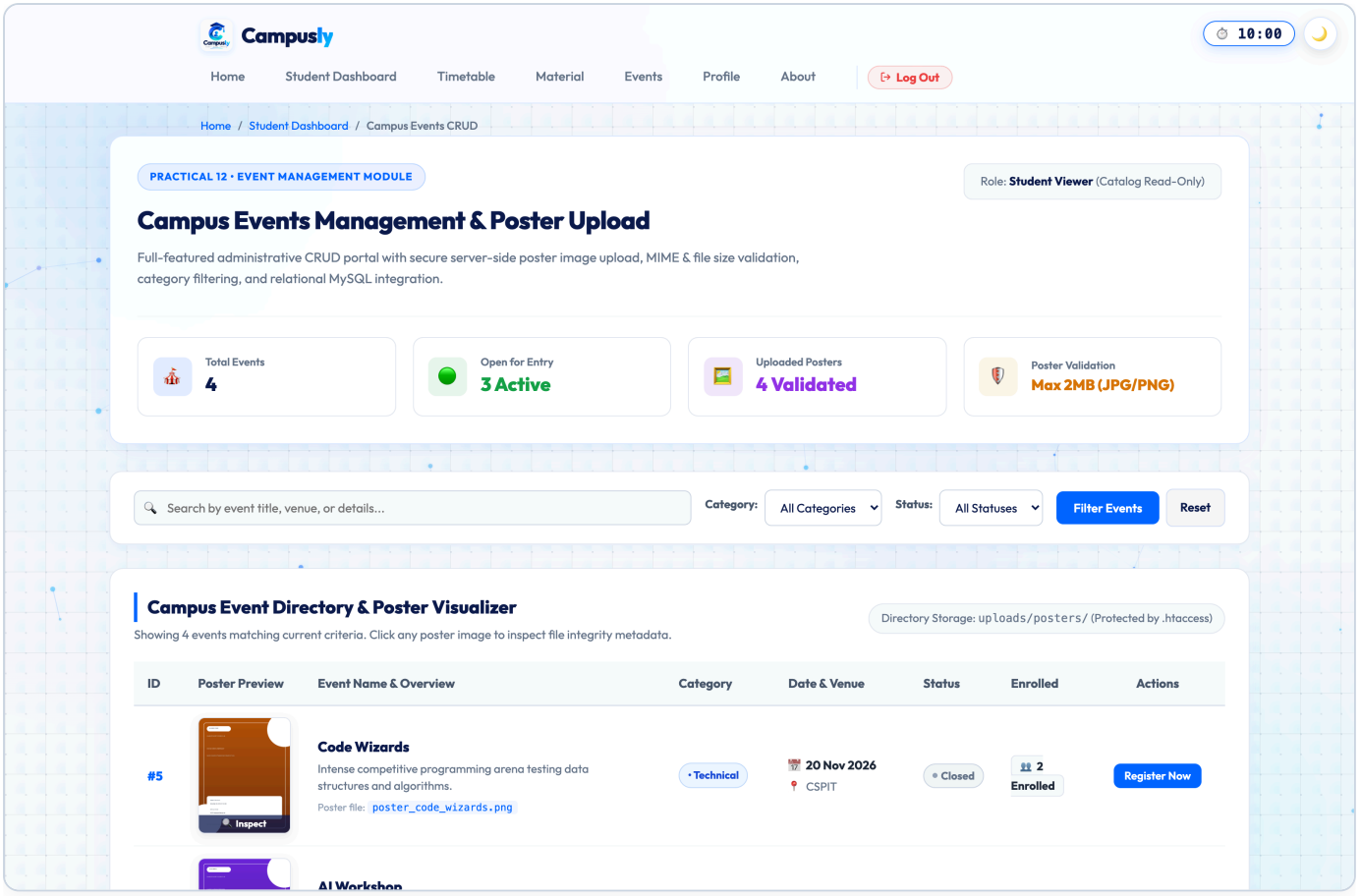
Real-time filtering: Typing search keywords ("Aditya") and selecting department ("CSE") instantly updates the records table via RESTful API queries without full page refresh.



7. Campus Events Catalog & Relational Registrations

7.1 Campus Events Catalog & Event Registration Junction

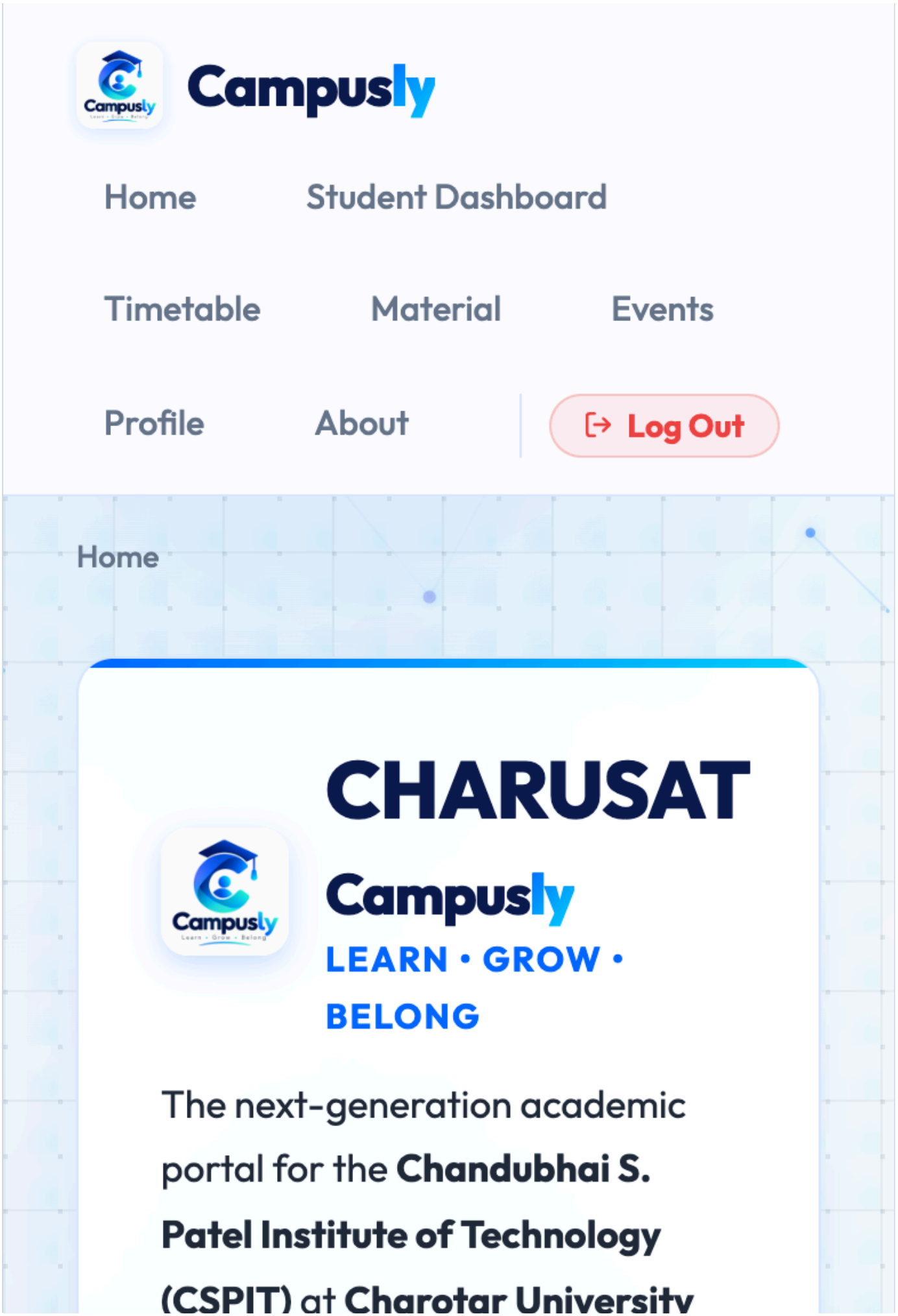
Campus events catalog (events.php) showcasing event posters (Hackathon, AI Workshop, Code Wizards), venue schedules, and live registration buttons linked via 3NF foreign keys.

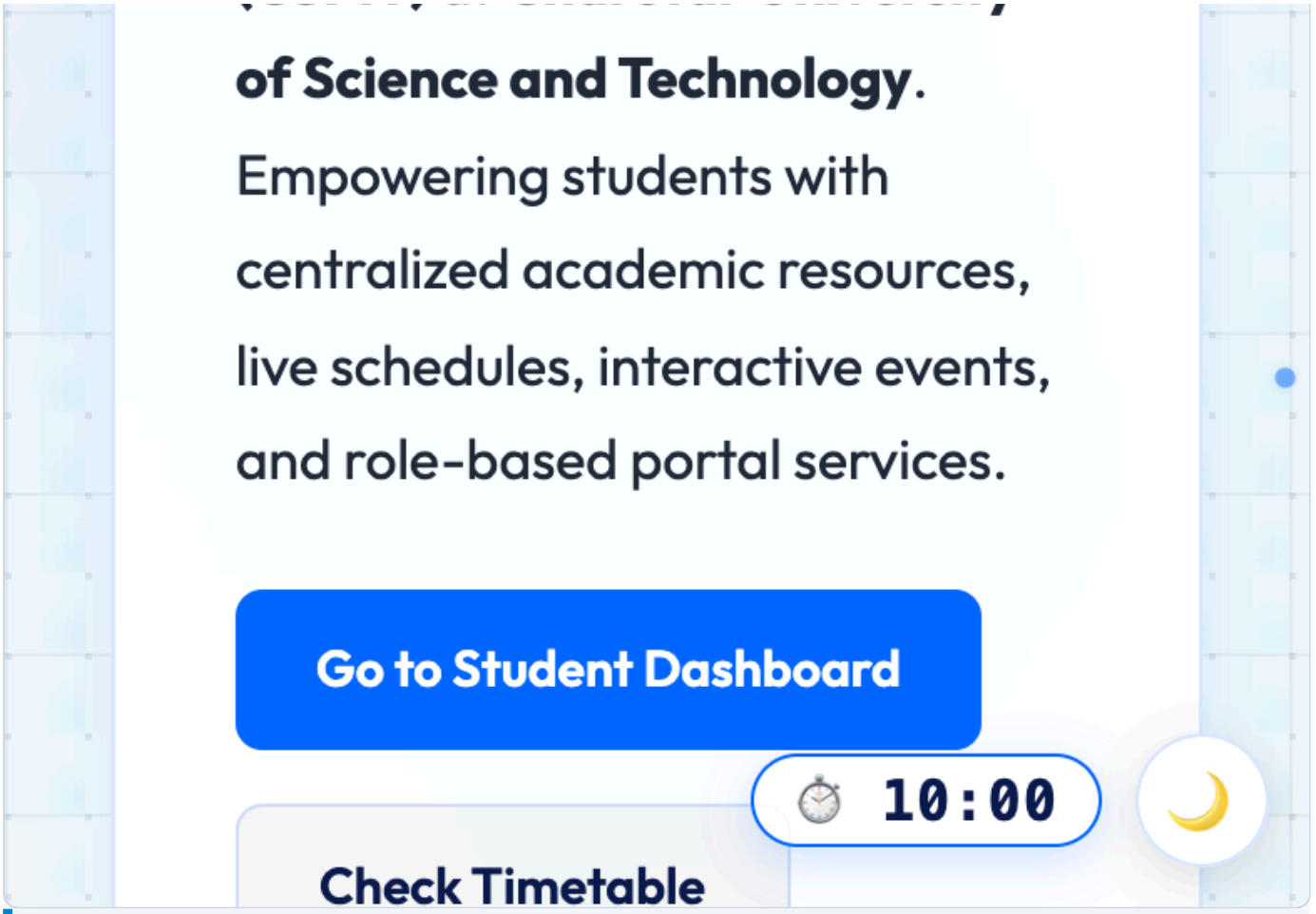

Campusly
10:00
Home
Student Dashboard
Timetable
Material
Events
Profile
About
Log Out
Home / Student Dashboard / Campus Events CRUD
PRACTICAL 12 • EVENT MANAGEMENT MODULE
Role: Student Viewer (Catalog Read-Only)
Campus Events Management & Poster Upload
Full-featured administrative CRUD portal with secure server-side poster image upload, MIME & file size validation, category filtering, and relational MySQL integration.
Total Events
4
Open for Entry
3 Active
Uploaded Posters
4 Validated
Poster Validation
Max 2MB (JPG/PNG)
Search by event title, venue, or details...
Category:
All Categories
Status:
All Statuses
Filter Events
Reset
Campus Event Directory & Poster Visualizer
Showing 4 events matching current criteria. Click any poster image to inspect file integrity metadata.
Directory Storage: uploads/posters/ (Protected by .htaccess)
ID
Poster Preview
Event Name & Overview
Category
Date & Venue
Status
Enrolled
Actions
#5
Inspect
Code Wizards
Intense competitive programming arena testing data structures and algorithms.
Poster file: poster_code_wizards.png
• Technical
20 Nov 2026
CSPIT
Closed
2 Enrolled
Register Now
AI Workshop

8. Responsive Mobile Accessibility & Viewport Layouts

8.1 Mobile Viewport Layout (iPhone 14 Pro / 390px Viewport)

Demonstration of mobile responsiveness: Fluid card stacking, accessible touch navigation, and responsive typography ensuring seamless access on smart devices.


Campusly
Home
Student Dashboard
Timetable
Material
Events
Profile
About
Log Out
Home
CHARUSAT
Campusly
LEARN • GROW • BELONG
The next-generation academic portal for the Chandubhai S. Patel Institute of Technology (CSPIT) at Charotar University



9. Local Deployment, Database Dump & GitHub Submission

9.1 Apache Virtual Host Local Deployment (apache_studenthub_vhost.conf)

Turnkey VirtualHost configuration for deploying StudentHub locally under the domain http://studenthub.local on Apache 2.4 / XAMPP with mod_rewrite activated.

Apache Virtual Host Local Deployment Configuration (`apache_studenthub_vhost.conf`) Intermediate Extension Verified

```
# ======================================================================
# Practical 15: Apache Virtual Host Configuration for StudentHub Portal
# System: StudentHub Academic Management Portal (CSPIT, CHARUSAT)
# Domain: http://studenthub.local
# Document Root: /Users/jaynavadiya07/Desktop/Project
# ======================================================================

# 1. macOS / Linux Apache Virtual Host Definition
<VirtualHost *:80>
    ServerAdmin admin@studenthub.local
    ServerName studenthub.local
    ServerAlias www.studenthub.local

    DocumentRoot "/Users/jaynavadiya07/Desktop/Project"

    <Directory "/Users/jaynavadiya07/Desktop/Project">
        Options -Indexes +FollowSymLinks +MultiViews
        AllowOverride All
        Require all granted

        # Priority Directory Index
        DirectoryIndex index.php index.html

        # URL Rewrite Engine for RESTful API and Clean Routing
        <IfModule mod_rewrite.c>
            RewriteEngine On
            RewriteBase /
            RewriteCond %{REQUEST_FILENAME} !-f
            RewriteCond %{REQUEST_FILENAME} !-d
            RewriteRule ^(.*)$ index.php [QSA,L]
        </IfModule>

        # Security Headers
        <IfModule mod_headers.c>
            Header always set X-Frame-Options "SAMEORIGIN"
            Header always set X-Content-Type-Options "nosniff"
            Header always set X-XSS-Protection "1; mode=block"
            Header always set Referrer-Policy "strict-origin-when-cross-origin"
        </IfModule>
    </Directory>

    # Custom Error Documents
    ErrorDocument 403 /dashboard.php
    ErrorDocument 404 /index.php

    # Access and Error Logs
```

9.2 MySQL studenthub Database Dump & Table Counts Verification

Database verification proving full 3NF relational integrity: Table counts confirm 25 Students, 4 Events, 14 Registrations, 4 Users, and 25+ Audit Logs.

MySQL Database Dump & Verification: `studenthub` Database

Practical 15 Relational 3NF Schema

```
$ mysql -u root -e "USE studenthub; SHOW TABLES; ... counts"

Tables_in_studenthub
audit_logs
event_audit_log
events
registrations
students
users
tbl     count
events  4
students        25
registrations   14
users   4
audit_logs      29
```

Database Dump File: **studenthub.sql** (217 lines, InnoDB, utf8mb4)

Status: **All Foreign Keys & Constraints Verified**

9. Local Deployment, Database Dump & GitHub Submission (Contd.)

9.3 GitHub Repository Submission & Clean Git History

Git terminal output confirming remote origin (git@github.com:cs-cspit/Student-Portal-25CS052.git), branch main up-to-date, and clear semantic commit history.

